Isolation Forest Anomaly Detection and Long Short-Term Memory Traffic Prediction

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***Abstract*— The integration of Artificial Intelligence (AI) with network traffic analysis offers a new approach to monitor and to predict network behavior. By leveraging Grafana, Prometheus and Node Exporter for real-time data collection and visualization, the system enhances traditional monitoring methods. These tools play a crucial role in improving network management: Prometheus collects and stores network data, while Node exporter delivers detailed metrics on endpoint devices and Grafana comes with an advanced visual interface, enabling efficient traffic analysis and monitoring. Machine Learning models were used for anomaly detection and traffic forecasting to identify potential issues and to predict the network conditions. By combining these technologies, the system enabled real-time monitoring, while also supporting anomaly detection and traffic prediction, thereby improving network performance and optimizing resource allocation.**

Keywords*— *artificial intelligence, Grafana, machine learning network traffic analysis, Prometheus.

I. Introduction

In the context of today's rapidly advancing technologies, the increasing reliance on networked systems has highlighted the need for advanced solutions that offer better monitoring, management, and prediction of network behavior [1], [2]. One such solution comes in the form of network traffic monitoring and analysis systems, which provide real-time insights into network performance [6]. Traditional network monitoring methods, while effective, often fall short in terms of proactive management and predictive capabilities [3]. To address these limitations, this project integrates Grafana, Prometheus, and Node Exporter for real-time data collection and visualization [4], combining them with Artificial Intelligence (AI) for anomaly detection and traffic prediction [2], [5]. By leveraging these technologies, the system enhances traditional monitoring by offering more precise forecasting, improving network performance, and optimizing resource allocation [1], [2]. The use of AI in this context enables the identification of potential issues before they escalate, offering a smarter, more dynamic approach to network management [3], [5]. This approach represents a significant step towards making network traffic analysis more predictive and efficient [2], [6].

This paper is structured as follows: Section II presents the proposed intelligent network monitoring architecture, which integrates Prometheus and Grafana with AI-based modules for traffic prediction and anomaly detection. Section III describes the implementation details and the experimental setup in GNS3 emulator. The paper ends with conclusions and future work.

II. Network monitoring architecture

The proposed architecture is divided into three sections, as in Fig.1.

- *Network topology* – represent the monitored topology which includes Docker containers that simulate real endpoints where metrics are transferred to the next section through a router [4].
- *Data Collection & Visualization* is the place where metrics are processed to obtain relevant graphs for monitoring desired network and these metrics are also passed to AI models' section to predict traffic patterns and detect anomalies [6], [7].
- Each *Docker container* has a Node Extractor server running on it which provides a comprehensive set of metrics about the status of system resources [6]. These metrics are essential for monitoring performance and detecting problems in network infrastructure [1].

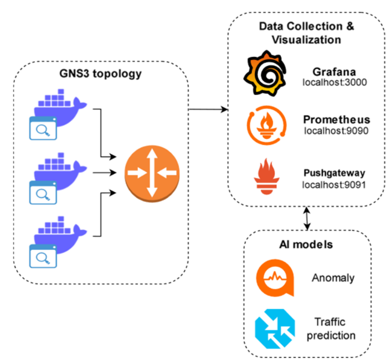

Fig. 1. Intelligent Monitoring System using Prometheus, Grafana, and AI

The Data Collection & Visualization component serves as the central point for aggregating and presenting network metrics. It is composed of three key tools: (a) Prometheus: a powerful time-series database used to collect and store metrics from network devices and systems [6]; (b) Pushgateway: a service that receives custom metrics (such as predictions and anomaly flags) pushed by external AI scripts; and (c) Grafana: an open-source analytics and visualization platform used to create dashboards that display these metrics stored in Prometheus in real time [7]. Prometheus scrapes metrics both from the Simple Network Management Protocol (SNMP) - enabled router and from Node Exporter running on the containers, while also pulling pushed data from Pushgateway [6]. Grafana then provides intuitive visual insights into system health, network performance, and potential anomalies [7]. The AI models section introduces advanced data-driven components designed to enhance traditional network monitoring. These models process the time-series collected by Prometheus to identify patterns, detect irregularities, and anticipate future behavior [1],[3]. This section includes two core functionalities: *Anomaly Detection:* a model trained on historical data to learn what constitutes normal behavior in the network(e.g. CPU usage, traffic flow, packet drops) [1],[3]; *Traffic Prediction:* a model that forecast future network load based on trends observed in past metrics, using DL models such as Long Short-Term Memory (LSTM) [2],[10].

III. Implementation and Experimental Results

The test environment uses GNS3 emulator to integrate virtual devices and Docker containers into a controlled, reproducible topology for both standard and AI-driven monitoring simulating real-world conditions [4].

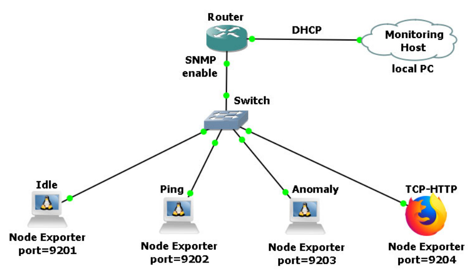

Fig. 2. GNS3-based network topology

The GNS3 topology links Docker containers (each running Node Exporter) to a Cisco router with SNMP enabled. It uses Dynamic Host Configuration Protocol (DHCP) to connect to the cloud (local PC) and it performs port forwarding, allowing Prometheus to access container metrics [4],[5]. On the PC, Prometheus, Pushgateway, and Grafana handle real-time metric collection, anomaly detection, and traffic prediction [6],[7]. To better illustrate the AI mechanisms see the following key code snippets: *Isolation Forest anomaly detection* (where -1 = anomaly and 1 = normal data).

```
model = IsolationForest(contamination=0.1)
df["anomaly"] = model.fit_predict(X)
```

LSTM traffic prediction:

```
model=tf.keras.models.Sequential([
tf.keras.layers.LSTM(32,activation='relu',input_sh
ape=(X.shape[1], 1)), tf.keras.layers.Dense(1)])
model.fit(X, y, epochs=5, verbose=0) y_pred =
model.predict(X[-1].reshape(1, X.shape[1], 1))
```

This builds and trains an LSTM neural network to learn traffic patterns and predict the next traffic value based on past data sequences [2], [10]. The traffic prediction graph (see Fig. 3) displays the system's forecasted network values over time. Negative values can appear due to model limitations or input noise, highlighting moments where predictions diverge and pointing to potential inaccuracies or unusual patterns[3]. The model runs automatically at set intervals, retraining from scratch each time on the latest data and generating new predictions, ensuring it constantly adapts to recent network behavior [1],[2].

Fig. 3. Traffic prediction over time

The automation mechanism built into this project plays a crucial role in ensuring continuous and up-to-date monitoring. Specifically, the system is designed to automatically execute both the anomaly detection and traffic prediction scripts at fixed time intervals (e.g., every 60 seconds) without requiring manual intervention. Each time the scripts run, they collect the latest metric data from Prometheus, retrain their respective models (Isolation Forest for anomalies, LSTM for predictions) on the most recent dataset, and then generate updated outputs. These outputs, anomaly flags and predicted traffic values, are automatically pushed to the Pushgateway, from where Prometheus picks them up for visualization in Grafana.

IV. Conclusions and Future Work

This project successfully demonstrates the integration of advanced AI techniques into a virtualized GNS3 network environment, combining real-time data collection, efficient metric aggregation, and dynamic visualization. By designing an automated system that continuously trains anomaly detection and traffic prediction models, the setup not only replicates realistic network behavior but also delivers actionable insights into system performance and potential issues. The use of Docker containers and port-forwarded SNMP metrics ensures modularity and flexibility, making the solution scalable for future extensions

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